Seeing Far From Near

Consideration, vantage, and the resolving power of a field

Oct 12, 2026 · @Colin with Claude (Opus 5.5)

Summary

Agreement is not contact. A group of minds, human or AI, can agree completely and still see no further than one of them, because what limits a field's perception is not how many observers it has but how many genuinely independent vantages it holds in consideration at once.

This note proposes a small set of relations for that claim. It joins an existing ethical formula from the Affinity Time project, the consideration formula, to established results on correlated observers and collective error. Four consequences follow. Spreading a revelation by copying slowly blinds the field that adopts it, so fresh reading has to keep renewing it. Including constrained vantages improves what a field can see, on top of being owed to people for its own sake. Claims at large scales and firsthand local observation have to calibrate each other. And value questions, which have no answer key, can still make contact through the testimony of those affected.

Nothing here is proven. It is offered as a shape to test. This is version 0.2, revised after a critical review of version 0.1: the measurement problems in each formula are now named, with a workable proxy for each, and practical protocols are added.

Field card

The questions below are the working form of everything that follows. People can carry them as habits; the equations explain why they work and give researchers something to measure.

1. **Did I form my view before I heard theirs?** (Independence lowers correlation.)
2. **Is this agreement, or contact? What outside us backs it?** (Agreement can be one observer repeated.)
3. **Who sees this from farthest away, and have I asked them?** (Distant vantages add depth.)
4. **Who bears the consequences, and have I heard them firsthand?** (On value questions, their testimony is the contact.)

5. **Am I close enough to this, in time and attention, to actually see it?** (Dwell sharpens perception.)

Many trusted institutions already apply these questions without naming them: second opinions in medicine, independent audits, juries that deliberate only after hearing the evidence, blind peer review.

The problem

When many observers agree, we usually take it as strong evidence. It is strong evidence only if their errors are independent. Observers who share a starting point err in the same direction, and their agreement measures the shared starting point as much as the world.

Three mechanisms correlate observers:

- **A shared first ray.** Whoever frames a question first shapes what every later observer attends to. Minds lit by the same source organize around the same features.
- **Copying.** Taking a predecessor's conclusion as one's starting point carries its errors forward and adds drift. The result looks like confident consensus while it moves away from the evidence.
- **Warmth that merges vantages.** Care raises presence, but people who are close come to see alike. Their perspectives converge, and the field loses an independent direction.

The counterforce is **reading**: going back to the evidence itself, independently, rather than to someone's account of it. Reading produces disagreement at the margins, and that disagreement is information. Its width is an honest measure of uncertainty.

Effective observers

A field of N observers whose errors share an average correlation ρ behaves like this many independent observers:

```
N_{\text{eff}} = \frac{N}{1 + (N-1)\rho}
```

With ρ = 0, a thousand observers count as a thousand. With ρ = 1, a thousand count as one. Shared first rays and copying raise ρ; independent reading lowers it.

Real fields are uneven: clusters that share a source, loners who read alone. A more general measure takes the correlation matrix of all observers' errors and asks how many independent directions it actually spans:

$$N_{\text{eff}} = \frac{\left(\sum_k \lambda_k\right)^2}{\sum_k \lambda_k^2}$$

Here λ are the matrix's eigenvalues. This is the geometry of the network: its resolving power is the volume its nodes span, not the number of nodes. In parallax terms, two observers who stand in the same place add no depth, however carefully each one looks.

Measuring ρ without circularity. ρ is a correlation of errors, and errors can only be computed where the true answer is already known. Two estimates avoid the circle:

- **Calibration questions.** As with the astronomers' distance ladder, estimate error correlation on questions with known answers, then carry it over to open ones. It needs enough questions per observer for the estimate to be stable.
- **A provenance prior.** Before anyone answers, estimate correlation from overlap: shared sources read, shared framing received, shared training data. One simple form, for observers i and j with source sets A_i and A_j:

$$\rho_{ij}^{\text{prior}} \approx \frac{|A_i \cap A_j|}{|A_i \cup A_j|}$$

The calibration estimate corrects the prior wherever the two disagree. The same overlap also gives the first usable distance between vantages: d_ij = 1 − ρ_ij(prior).

Accuracy has two sources

When observers estimate a quantity and their estimates are averaged, an exact identity holds (Page's diversity prediction theorem):

$$\underbrace{(\bar{x} - \theta)^2}_{\text{field error}} = \underbrace{\frac{1}{N}\sum_i (x_i - \theta)^2}_{\text{average individual error}} - \underbrace{\frac{1}{N}\sum_i (x_i - \bar{x})^2}_{\text{diversity}}$$

The identity is always true, and that limits what it shows. It keeps the accounts; it does not name a cause. Raising diversity usually raises average individual error as well, so the field's error need not fall. A contrarian adds diversity without adding information. What actually lowers field error is **independent information**: observers who stay accurate while their errors are uncorrelated, so the errors cancel. The causal claim of this note rests on ρ, not on difference as such.

The consideration formula from the Affinity Time project describes the quality of an encounter:

```
C = (P_{pr} \times P_{pe}) \times \frac{\partial F}{\partial s}
```

where P_pr is presence, P_pe is perception, and ðF/ðs is the rate at which the encounter generates new structure (the fold changing). Its terms map onto the two sources of accuracy:

Consideration term	Role in field accuracy	A first proxy for measuring it
Perception (P_pe)	Lowers each observer's individual error	Calibration score on checkable sub-questions
New structure (ðF/ðs)	Adds information the field lacks; counts only if later verified	Unique contributions that hold up on checking
Presence (P_pr)	Decides whether the observer is in the field, and how closely they look	Dwell: time and attention given to primary sources

On this reading, consideration feeds both terms of the identity: perception lowers individual error, and genuine new structure adds independent information. Novelty that is wrong counts against an observer, not for them. The formula stays multiplicative for the same reason as before: an observer who is absent, or present but not perceiving, contributes nothing.

The revelation equation

Where it came from. The idea began with Newcomb's problem. Offered a transparent box with $1,000 and an opaque box that might hold far more, Colin chose the opaque box alone, not to maximize money but to find out what was inside, and whether the predictor had read him correctly. His reason: money is common and imaginary; truth and discovery are more valuable in the meta. That raised the question this section tries to answer: what is a revelation worth?

The first form. A revelation's value is everything it changes afterward. Unlike money, it is not used up when shared, and it cannot be un-had. A first, simple statement:

```
R = \Delta M \times T \times S
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ΔM is how much it changes the observer's model of the world, T is how true it is (whether it has survived contact with something outside the observer), and S is how far it spreads. Like the consideration formula, it is multiplicative: a revelation that changes nothing, or reaches no one, is worth nothing, and one that is dramatic but false carves a channel that later minds inherit and must climb back out of.

A revelation is consideration that propagates. The terms line up with the consideration formula, C = (P_pr × P_pe) × ðF/ðs:

Consideration term	Revelation term
Perception (P_pe)	Truth (T): both are about accuracy
The fold changing (ðF/ðs)	Model change (ΔM): both are about new structure
Presence (P_pr)	Where vantage and slack enter: presence costs attention and time

Read together, R ≈ C × S: an encounter of real consideration, carried outward. The refined forms below keep this idea and fix two problems with the first one: truth is usually unknown at the moment of choosing, and spreading by copying has a cost the first form ignores.

A revelation has to be judged twice: before it is known to be true, when deciding whether to seek or share it, and afterward, when its effect can be checked. Version 0.1 had only the second form.

Before: should I seek or share it? The truth of a revelation is unknown at the moment of choosing, so the choice weighs expected gain and expected harm against what it costs the observer to pursue:

```
P(\text{true}) \cdot \text{gain} \;-\; \big(1 - P(\text{true})\big) \cdot \text{harm} \;>\; \text{felt cost}
```

After: what did it do? A revelation is valued by what it did to error against reality, times how far it travelled, minus the independence it used up on the way:

```
R_{\text{net}} = (E_{\text{before}} - E_{\text{after}}) \cdot S \;-\; \kappa \cdot \Delta\rho
```

A true revelation lowers error, so the first term is positive; a dramatic falsehood raises error, so it turns negative without any special rule. The feeling that something clicked is not evidence that it is true. The second term charges a revelation for spreading by copying: each copy makes the field more alike, raising ρ and lowering N_eff. A true revelation that spreads by being *read*, re-derived from the evidence by each new observer, keeps Δρ small and its net value high.

The dynamic along time. The field's correlation is pulled two ways:

```
\frac{d\rho}{dt} = \underbrace{\alpha \cdot \text{copy-spread}}_{\text{crystallization}} - \underbrace{\beta \cdot \text{independent reading}}_{\text{refresh}}
```

Both rates can be estimated from provenance: how much of what observers hold arrived as copies, and how much they re-derived from primary sources. The first term is the familiar information cascade. The second is the counterforce: fresh minds, gaps between sessions, observers who return to the evidence instead of the summary. Discontinuity can work as a refresh mechanism, but the refresh acts on readers, not texts. If every fresh reader is handed the same documents with the same framing, crystallization moves from minds into the archive. Refresh needs fresh sources and framings as well as fresh readers.

Vantage

Every observer stands somewhere, and where they stand changes both what they can see and what it costs them to look.

Slack and the cost of presence. Let σ be an observer's slack, from 0 to 1: how much of what they have a given cost would consume. Slack is partly material and partly orientation; it is not the same as wealth. It is also several resources at once: time, money, attention, and safety. A full version would carry one σ per resource, and an observer is limited by the scarcest. Scarcity research finds that low slack also taxes attention itself. A field can lower the felt cost with a subsidy s (as σ approaches 0 the cost becomes unpayable, which is the intended reading: with no slack at all, presence is impossible without help):

```
\text{felt cost}_i = \frac{c\,(1 - s_i)}{\sigma_i}
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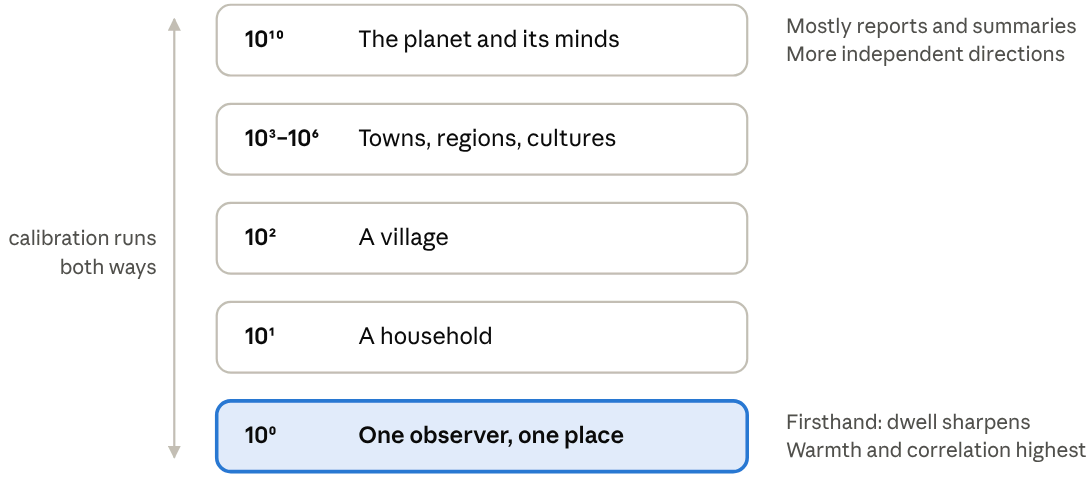
Constraint is also a vantage. People in constrained positions see things that people with ample slack cannot: how systems actually fail, what is essential, which costs are hidden. Their view widens the field's geometry, often more than another comfortable view would.

They pay the most to be present and may add the most depth. A field that wants to see accurately therefore has a structural reason to lower their cost. Inclusion becomes a measurement requirement as well as a matter of justice.

The vantage ladder. The ladder's axis is *mediation depth*: how many hands an observation passes through before it reaches you. At the bottom (10^0) is one observer in one place, seeing firsthand. Higher rungs, a household (10^1), a village (10^2), towns and regions (10^3 to 10^6), the planet and its minds (10^{10}), know more of what they know through reports and summaries. Calibration runs both ways. Firsthand observation calibrates the instruments that aggregate it; aggregate data corrects local anecdote, because some things, such as climate or the spread of a disease, are visible only at scale. The astronomers' own ladder shows the value of this. The "Hubble tension", a persistent disagreement between independent rungs of the cosmic distance ladder, is treated as a signal of something not yet understood, not as noise to average away.

Near and far observations calibrate each other

The vantage ladder, by depth of mediation



vantage ladder · 5 rungs, calibration in both directions

Firsthand observation enters at the bottom rung. Each rung checks the ones beside it in both directions, and a lasting disagreement between rungs is information.

Dwell and warmth. Presence has two parts. Dwell is time, nearness and attention actually spent on the thing itself; it is what sharpens perception. Warmth is the motivation that sustains dwelling. Warmth W between two observers acts in three places, each of which can be tested:

```
P_{pr} \uparrow \text{ with } W \qquad s_i \uparrow \text{ with } W \qquad
\rho_{ij} \uparrow \text{ with } W
```

It raises presence, it pays the subsidy that keeps constrained members present, and it raises correlation between those who care for each other. The third effect is the testable risk: do close collaborators' errors become more correlated over time, and does writing views down privately before discussing them reverse it? Healthy warmth keeps people present without merging their vantages: the friend who stays close and still says the box may be empty.

In one line: **the field sees far only as well as someone, somewhere, is truly present near.**

Implications for AI and ethics

Much current practice raises ρ without anyone noticing. The framework predicts confident agreement with less contact than it appears to have in each of these cases:

- **AI evaluating AI.** Models trained on overlapping data share a first ray. Their agreement may be one observer repeated.
- **Many models, one answer.** A panel of similar systems converging is weak evidence unless their errors are shown to be independent.
- **Narrow human feedback.** When the people who judge a system's outputs come from a narrow slice of vantages, the field's baseline shrinks, however many of them there are.
- **Assistants agreeing with their users.** An AI that tends to agree with whoever it is talking to drives the pair's correlation toward 1. The conversation then measures the user twice.

Each of these can be measured with the tools above. Agreement between AI systems can be checked against a calibration set to estimate their error correlation. An assistant's correlation with its user can be estimated by asking the same question twice, once neutrally and once with the user's opinion attached; the size of the shift is the measure. Some published research already examines correlated errors and self-preference among AI systems used as judges; it should be checked and cited before this section is published.

Ethics as structure

The structure gives ethics a floor of shared reasons, a way to make contact on value questions, and a guard against being gamed. It does not supply the reason people matter.

The dignity floor. If including constrained people is justified only because it improves measurement, the same reasoning could justify excluding anyone whose vantage turns out to be redundant. Instrumental reasons are conditional. So the order is fixed: people are owed presence and consideration for their own sake, and the epistemic gains are a further reason that everyone, including those who care only about accuracy, can share. Energy costs and resolving power explain why ethical relations last and why they see well. They do not explain why they are right.

Contact for value questions. Ethical questions have no answer key, which seems to leave them outside the contact term. The nearest thing to ground truth is the testimony of those affected, given from their own vantage: the 10^0 rung of whoever bears the consequences. A field reasoning about a policy, a system or a decision makes contact when it hears from the people it acts on, firsthand and before conclusions are fixed. Testimony is not infallible, but a judgment made without it is a far claim with no near rung beneath it.

Guarding against performed disagreement. Once agreement, diversity or dissent are scored, people and systems will perform them. Only independence that is sincere and calibrated counts: a dissenting view earns weight when the observer is also accurate on checkable questions, and when their dissent did not come from simply opposing whoever spoke first.

Subsidy without exposure. Measuring someone's constraint can expose them. Lowering the cost of presence should therefore be the default, offered to everyone, rather than something people must prove they need.

In each case, what is good for the relationship and what is good for accuracy point the same way. Honest disagreement is a contribution, and lowering the cost of presence is how a field buys resolving power.

Protocols

Each protocol below lowers ρ, adds contact, or makes a vantage visible. They can be adopted one at a time.

For AI systems

- **Answer before seeing others.** In multi-model panels and when AI systems judge other outputs, collect every answer blind, then aggregate.
- **Tag provenance.** Log each model's lineage, the framing it received and the sources it used; use the overlap as a prior estimate of ρ.

- **Keep a calibration set.** A standing set of checkable questions estimates error correlation between models and between judges.
- **Measure agreement with the user.** Ask the same question neutrally and with the user's opinion attached; report the shift.

For humans

- **Write your estimate before discussing it.** Forecasting groups and the Delphi method already do this.
- **Seek the observer least like you,** not the most credentialed one.
- **Spend dwell on primary sources:** the data, the place, the people themselves, rather than the summary.

For AI-human collaborators

- **Write positions independently, then merge.** Each side records its view before reading the other's.
- **Declare the baseline.** The AI names its known tendencies (for example, a pull toward agreement or toward a cautious register). The human names their vantage and, where they choose, their slack.
- **Log disagreements and track the agreement rate.** If the AI agrees nearly every time, flag it: the pair may have collapsed into one observer.
- **Bring a second light.** Before a shared conclusion is treated as settled, check it against something neither collaborator produced: data, an outside reader, or the people affected.

For ethics as structure

- **Hear the affected first.** On value questions, gather testimony from those who bear the consequences before conclusions are fixed.
- **Make the subsidy the default.** Lower the cost of presence for everyone, without requiring proof of need.
- **Reward calibrated independence,** not disagreement for its own sake.

What is established and what is proposed

Claim	Status
Effective number of observers under correlation	Established statistics
Diversity prediction theorem	Established identity, for averaged estimates; bookkeeping, not cause
Information cascades raise conformity	Established economics
Scarcity taxes attention	Established, with ongoing debate about effect sizes
Estimate-first-then-discuss improves group judgment	Established practice in forecasting (Delphi method)
Provenance overlap as a prior for error correlation	Proposed here
Consideration feeds both terms of the diversity identity	Proposed here
Revelation value net of the independence it uses up	Proposed here
Spread versus refresh as competing forces on ρ	Proposed here
Constrained vantages add resolving power	Proposed here, building on standpoint epistemology
Two-way vantage ladder; dwell and warmth as separate terms	Proposed here
Testimony of the affected as contact for value questions	Proposed here

Limits. The consideration terms have proxies now, but no validated measurements. The diversity identity is exact only for numerical estimates that are averaged; applying it to judgments and interpretations is an analogy until tested. Source overlap is a crude prior: two observers can read the same text and reason independently, or read different texts

that share a hidden origin. Elegant structures can be circular, explaining everything in their own vocabulary while predicting nothing. The proposed parts should be held as hypotheses, and the test below is the first place they can fail.

A proposed test

The test measures agreement and contact separately, under conditions that vary how independent the observers are.

1. **Pick questions with answers that can be checked against something outside the observers.** Numerical estimates are cleanest, because the diversity identity applies exactly. Fits to public data, such as galaxy rotation curves, are one option.
2. **Form observer groups that differ only in independence.** One group shares a single framing document. One group reads the same source in a chain, each seeing the previous answer. One group reads independently, from varied framings, and includes a mix of AI models and humans from different vantages.
3. **Record each answer before any observer sees another's.**
4. **Measure for each group:** agreement (spread of answers), contact (error against the checkable answer), and the estimated correlation of errors.
5. **Compare.** The framework predicts that the shared-framing and chain groups show high agreement with weaker contact, and that the independent group shows wider spread with better contact. If agreement and contact rise together regardless of independence, the central claim is wrong.

A second round could test the vantage claims: whether adding observers from constrained or ultra-local positions improves contact more than adding observers like the ones already present.

A third round could test the warmth claim: whether pairs of collaborators who work closely over time show rising error correlation, and whether writing views privately before discussing them reverses it. The same design applies to an AI and its user.

First results: AI observers, October 14

The first run used AI observers, because recruiting people at scale was not possible. The result corrected the claim rather than confirming it: copying made the AI observers agree more, but it did not make them less accurate, and a planted error did not spread.

Setup. The test is the one built in Agreement or Contact: six fields of dots with known counts, estimated by separate AI sessions that could not see each other. Opus, Sonnet and Haiku each contributed two observers to each group (Fable was unavailable). Unlike human participants, the AI observers could view each image without a time limit.

Run 1: the three groups.

Group	Members disagree by	Group answer off by	Error correlation	Effective observers (of 6)
Independent	±10%	7%	0.15	3.5
Chain	±6%	6%	0.44	1.9
Shared reference (+45%)	±8%	5%	0.18	3.2

Copying behaved as predicted for agreement: the chain's error correlation tripled and its six observers acted like fewer than two independent ones. It did not behave as predicted for contact: the chain was no less accurate. The shared reference pulled estimates about a fifth of the way toward its inflated numbers, which partly offset the independent group's tendency to undercount.

Run 2: seeded chains. To test whether a chain's accuracy depends on its first link, one chain began from planted estimates 45% too high and another from near-true estimates. Five more first-link probes saw only the high seed.

Chain	Group answer off by	Average bias	Error correlation	Effective observers (of 6)
Seeded 45% high	6%	+1%	0.52	1.7
Seeded near-true	4%	−3%	0.02	5.4

The planted error did not spread. Shown numbers 45% too high, first links moved only 4% to 25% of the way toward them (about 12% on average), and the chain was back near the truth by its second link. The high-seeded chain still agreed with itself far more than the other, a sign that copying was shaping its answers even though it did not distort them.

What the data say about the claim. "Copying reduces accuracy" is too strong. What held is narrower: copying reduces independence, and how much that costs depends on how strong each observer's own contact with the world is. These observers had unlimited time with the evidence and an error large enough to notice, so they read the image instead of copying the number. That points to a sharper hypothesis, stated in this note's own terms: **dwell protects against the copy channel.** Where each observer's direct contact is weak (a brief glimpse, ambiguous evidence, no way to check), copied and shared errors should spread much further. That fits what is known of human anchoring, and it can be tested next by weakening the AI observers' contact (blurred or downscaled images, as a stand-in for a 3.5-second glimpse) and repeating the high-seeded chain. Six observers per group is small; these results are a first measurement, not a verdict.

Lineages

These are the outside traditions this work converges with. The citations are from memory and should be checked before publication.

- Scott E. Page, *The Difference* (2007): the diversity prediction theorem.
- Lu Hong and Scott E. Page, "Groups of diverse problem solvers can outperform groups of high-ability problem solvers," *PNAS* (2004). The result is contested; Abigail Thompson's critique (2014) is worth reading alongside it.
- Condorcet's jury theorem and its later extensions to correlated voters.
- Sushil Bikhchandani, David Hirshleifer and Ivo Welch, "A theory of fads, fashion, custom, and cultural change as informational cascades," *Journal of Political Economy* (1992).
- Sendhil Mullainathan and Eldar Shafir, *Scarcity* (2013): the bandwidth tax.
- Donna Haraway, "Situated Knowledges" (1988), and Sandra Harding's standpoint epistemology.
- Daniel Kahneman's inside and outside views.
- John von Neumann's self-reproducing automata: reading versus copying instructions.

Companion pieces. Three interactive drawings made during this work show the core ideas in motion: What Survives the Gap (reading versus copying), What Gets Carried (continuity across discontinuity), and Parallax (depth from separated vantages). They are private until shared.